

the Nexon EV and Tigor EV offerings powered the company's Ziptron electric drive technology. The Tigor EV is currently one of the most affordable four-wheel EVs on the market with prices starting from ₹12.49 lakhs. The Tigor EV is ARAI-certified for a 306 km range on a full charge.

The Nexon EV and the Nexon EV Max are the premium offerings from Tata Motors with the Nexon EV Max rated by ARAI to last for 437 km on a full charge thanks to its large 40.5 kWh battery. Tata Motors claims that the Nexon EV Max can do 0-100 km in less than 9 seconds with a top speed of 140 km/h. Pricing for the Nexon EV Max starts from ₹17.74 lakhs.

Mahindra Electric

Mahindra offers a variety of three-wheeler and four-wheeler EVs for the Indian market. Mahindra Electric's portfolio includes the eVerito sedan, e2O Plus compact car, eAlfa Mini auto-rickshaw, Treo electric auto, and the eSupro passenger and cargo van.

The eVerito is rated for a 181 km range, which can be extended via the company's Revive technology that allows the drivetrain to offer another 8 km of travel when the charge gets depleted. The eVerito comes in D2 and D6 variants and starts from ₹10.11 lakhs.

The e2O Plus is a hatchback offering that caters to regular city commute. Available in three variants — P2, P4, and P6 — the Mahindra e2O Plus is rated to offer up to a 140 km driving range with a Revive distance of 5 km, up to a 15 kWh battery, and a fully automatic transmission. The e2O Plus starts from ₹7.48 lakhs for the P4 variant.

Apart from the above, a host of other players — both big and small — dot the Indian EV space. These include TVS's iQube scooter, Ola Electric's S1 and S1 Pro e-scooters, Simple Energy's Simple One, Hero Electric's AE 47, and Bajaj's Chetak Electric among others.

Some of the upcoming EVs in India include the Tata Altroz EV, Mahindra eKUV100, MG eZS, Tesla Model S, Audi e-Tron, Ford Aspire EV, BMW i4, Kia EV6, and many others.

INCREASING EV ADOPTION IN INDIA

Though the steadily increasing number of EV players seems promising, a lot still needs to be done both at the governmental and individual OEM levels to boost EV penetration on Indian roads.

Impetus from the government

The Government of India has rolled out several schemes and allocated large budgetary provisions to bolster EV adoption. For instance, the Faster Adoption and Manufacture of Hybrid and Electric Vehicles (FAME) II scheme launched in 2019 comes with a ₹10,000 crore outlay to support EV manufacturing.

There's also a new Production Linked Incentive for Advanced Chemistry Cell Battery Storage (PLI-ACC) scheme that was announced in 2021 with an outlay of ₹18,000 crore for promoting local and global investments in battery technology.



Government think-tank Niti Aayog projects massive savings in energy for FAME II-eligible vehicles. According to Niti Aayog's analysis, FAME II-eligible vehicles will be able to cumulatively save 5.4 million tons of oil equivalent (MTOE) over their lifetime, which amounts to a massive ₹17.2 thousand crores of savings, 170 petajoules of energy reduction, and 7.4 million tons of CO2 emissions reduction over the lifetime of these vehicles.

If India were to indeed achieve an EV penetration 30% private cars, 70% commercial cars, 40% buses, and 80% two and three-wheelers by 2030, the think-tank estimates that this will lead to savings of 474 MTOE worth ₹1,521

thousand crores, a net reduction of 14 exajoules of energy, and reduction of 846 million tons of CO2 emissions over the lifetime of these vehicles.

As per estimates, less than 10% of the subsidies earmarked under FAME II have been used so far, indicating that there is still a lot of untapped potential and lots of scope for new and established vendors.

Apart from the Union Government's initiatives, most states and union territories have developed or are in the process of formulating their own EV policies. For instance, Karnataka aims to transition 100% of its three and four-wheeler goods vehicles to electric by 2030 while offering incentives for the first 100 entities who setup fast charging stations.

Maharashtra aims to generate an investment of ₹25,000 crores for EV manufacturing while providing jobs for 100,000 people.

Andhra Pradesh plans to have one million EVs by 2024 along with 100,000 fast and slow charging stations by that time. The state also wants to ensure that all government vehicles and ambulances become fully electric by 2024 and even has plans to stop registrations of ICE vehicles in the upcoming capital city by that time.

The National Capital Territory of Delhi projects 25% of new vehicle registrations to be electric by 2023 along with 50% of public transportation. The region's draft electric policy also envisions reuse and recycling policy for ageing EV batteries.

Initiatives from the industry

While governments try to incentivise EV adoption, the industry and consumers alike need to recognise the potential and advantages of promoting this industry.

Balancing pricing and features:

EVs continue to demand a premium over their ICE counterparts, and this is likely to continue for some more time until the economies of scale are attained. When customers pay upwards of ₹1 lakh for a two-wheeler EV, they do expect a premium experience both with the product and with after-sales service. Though subsidies help, the cost of an EV will continue to